Derivatives

1. Motivation

We usually have to optimize (maximize or minimize) some quantities. For example: Maximizing the profit, Maximizing Revenue; Minimizing Cost; Minimizing Risk and so on.

Derivative is a tool that will help to maximize/minimize certain quantities.

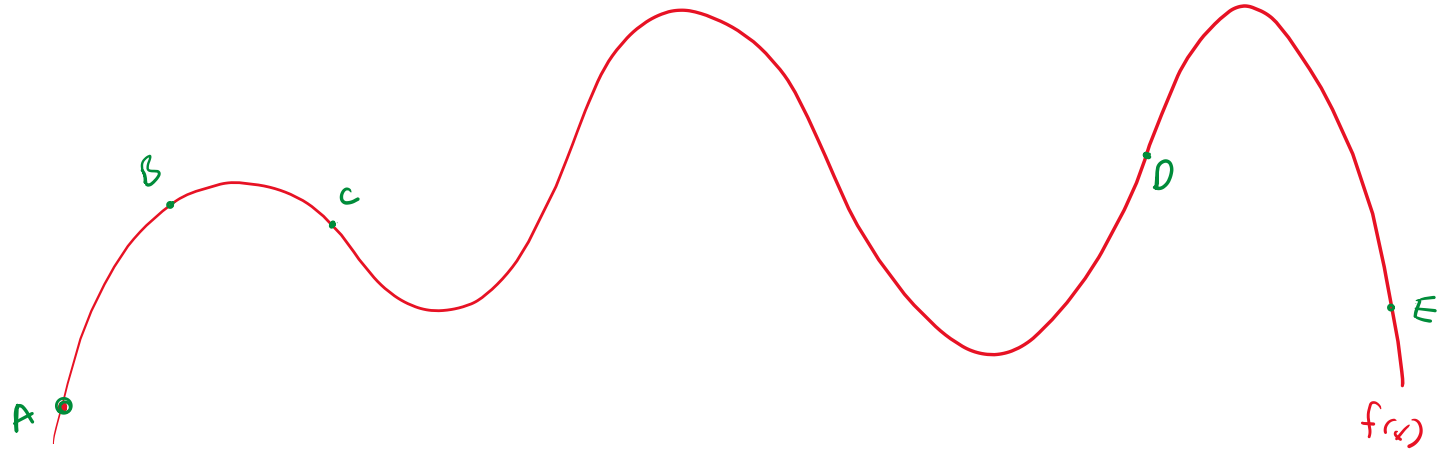
What is derivative?

2. Derivative of a function at a point.

The derivative of $f(x)$ at $x = a$ is a number denoted by $f'(a)$ [reads: f prime of a] measuring the "<u>changes</u>" of $f(x)$ at $x = a$.

Specifically:

1. If $f'(a) > 0$ then $f(x)$ goes up at $x = a$
2. If $f'(a) < 0$ then $f(x)$ goes down at $x = a$
3. The "speed" of going up / down is $|f'(a)|$



3. Calculating derivatives.

We will use several "Rules" to calculate derivatives.

<u>Rule 1</u> : Constant Rule

If $f(x) = c \Rightarrow f'(x) = 0$

<u>OR</u> : $(c)' = 0$

<u>Rule 2</u> : Identity rule

If $f(x) = x \Rightarrow f'(x) = 1$

<u>OR</u> : $(x)' = 1$

<u>Rule 3</u> : Power Rule :

If $f(x) = x^n \Rightarrow f'(x) = n \cdot x^{n-1}$

<u>Example</u>: Find $f'(x)$. Is $f(x)$ going up or down when $x = 1$?

① $f(x) = x^{10}$

$f'(x) = 10 \cdot x^{10-1} = 10x^{9}$

$f'(1) = 10 \cdot 1^{9} = 10 > 0$

$\Rightarrow$ $f(x)$ goes up when $x = 1$

② $f(x) = x^{2025}$

$\Rightarrow$ $f'(x) = 2026\, x^{2025}$

③ $f(x) = x^{7.5}$

$\Rightarrow$ $f'(x) = 7.5 * x^{6.5}$

④ $f(x) = \dfrac{1}{x^{10}} = x^{-10}$

$\Rightarrow$ $f'(x) = -10 \cdot x^{-10-1}$

$$= -10\, x^{-11}$$

$$= \frac{-10}{x^{11}}$$

<u>Rule 4</u> : Scalar Multiplication

$$\left(k \cdot f(x) \right)' = k \cdot f'(x)$$

<u>Example</u> :

① $f(x) = 10 \cdot x^3$

$$f'(x) = \left(10\, x^3\right)' = 10 \cdot \left(x^3\right)'$$

$$= 10 \cdot 3 \cdot x^2$$

$$= 30\, x^2$$

② $f(x) = 7x$

$$f'(x) = (7x)' = 7(x)' = 7$$

③ $f(x) = \dfrac{9}{x^4}$

$$\Rightarrow f(x) = 9 \cdot x^{-4}$$

$$\Rightarrow f'(x) = \left(9 \cdot x^{-4}\right)' = 9 \cdot (x^{-4})'$$

$$= 9 \cdot (-4) \cdot x^{-4-1}$$

$$= -36 \cdot x^{-5}$$

$$= \frac{-36}{x^5}$$

<u>Practice</u>: Find $f'(x)$. Is $f(x)$ going up or down when $x = 2$.

① $f(x) = x^{100}$

② $f(x) = 3x^{100}$

③ $f(x) = \dfrac{1}{x^7}$

④ $f(x) = \dfrac{9}{x^7}$

⑤ $f(x) = 6x$

⑥ $f(x) = -9x^3$

(5) Additions and Subtraction Rules.

$$[h(x) + g(x)]' = h'(x) + g'(x)$$

$$[h(x) - g(x)]' = h'(x) - g'(x)$$

<u>Example</u>: Find $f'(x)$

(1) $f(x) = x^{10} + 7x - 4$

$$f'(x) = (x^{10} + 7x - 4)'$$

$$= (x^{10})' + (7x)' - (4)'$$

$$= 10x^{9} + 7$$

(2) $f(x) = 9x^{4} - 6x^{2} + 9x + 3$

$$f'(x) = 36x^{3} - 12x + 9$$

Rule 6 : Product Rule

$$\left[f(x) \cdot s(x)\right]' = f'(x) \cdot s(x) + f(x) \cdot s'(x)$$

Example :

① $f(x) = (x+1) \cdot (x^2+1)$

$$f'(x) = (x+1)' \cdot (x^2+1) + (x+1) \cdot (x^2+1)'$$

$$f'(x) = 1 \cdot (x^2+1) + (x+1) \cdot (2x)$$

$$= x^2 + 1 + 2x^2 + 2x$$

$$= 3x^2 + 2x + 1$$

② $f(x) = (x^3 + x^2 + 1) \cdot (4x + 9)$

$$\Rightarrow f'(x) = (x^3+x^2+1)' \cdot (4x+9) + (x^3+x^2+1) \cdot (4x+9)'$$

$$= (3x^2 + 2x) \cdot (4x+9) + (x^3+x^2+1) \cdot (4)$$

Quotient Rule

$$\left(\frac{Top}{Bot}\right)' = \frac{(Top)' \cdot Bot - (Top) \cdot (Bot)'}{(Bot)^2}$$

Example :

① $f(x) = \dfrac{x+1}{x+2}$

$$f'(x) = \frac{(x+1)' \cdot (x+2) - (x+1) \cdot (x+2)'}{(x+2)^2}$$

$$= \frac{1 \cdot (x+2) - (x+1) \cdot 1}{(x+2)^2} = \frac{x+2-x-1}{(x+2)^2}$$

$$= \frac{1}{(x+2)^2}$$

Practice : Find $f'(x)$

① $f(x) = x^{10} + 6x^3 - 7x$

② $f(x) = (x^2 + x) \cdot (x^3 - 3)$

③ $f(x) = \dfrac{x^2+1}{x^2+3}$